

Introduction to AWS Cloud Services

A Practical Overview of Core AWS Services Used in Industry

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What is Cloud Computing ?

Cloud computing is renting computer power and storage over the internet instead of buying physical hardware. Instead of setting up expensive servers, wires, and hard drives in your own office, you connect to a provider like Amazon Web Services (AWS) to use theirs instantly. You only pay for the exact resources and seconds you actually use.

Traditional (On Premises)	Cloud (AWS)
You buy physical servers	Rent servers over the internet
You manage all hardware	AWS manages all hardware
Pay upfront — huge cost	Pay only for what you use
Hard to scale up or down	Scale in minutes, not months
Your team handles failures	AWS handles failures for you
Takes weeks to set up	Ready to use in minutes

AWS #1

Cloud Provider
Worldwide

AWS 200+

Services &
Tools Available

2006

Year AWS
Launched

What is AWS ?

- **Amazon Web Services (AWS)** is the world's most comprehensive and broadly adopted cloud platform — offering 200+ services from data centers globally. Launched in **2006**, AWS lets any organization rent computing, storage, databases, and more over the internet — paying only for what they use.

Before AWS

01

High Upfront Infrastructure Costs

02

Limited Scalability

03

Long Time Deployment

04

Complex Disaster Recovery

05

Global Reach Was Difficult

After AWS

01

No Upfront Costs

02

Instant Scalability

03

Faster Deployment

04

Global Reach with a Few Clicks

05

Built-in Disaster Recovery & High Availability

AWS holds approximately 31% of the global cloud market — ahead of Microsoft Azure (22%) and Google Cloud (10%)

Pay-As-You-Go



How to access Aws Services

01

AWS Management Console

Point-and-click web interface

The easiest way to get started. Open your browser, log in to console.aws.amazon.com, and manage every service through a visual interface. Perfect for beginners.

02

AWS CLI

Command Line Interface

Run AWS commands from your terminal. Type commands like `'aws s3 cp myfile.txt s3://mybucket/'` to interact with services quickly. Preferred by developers for automation.

03

AWS SDKs

Software Development Kits

Access AWS from inside your code. Available for Python (boto3), JavaScript, Java, .NET, and more. Used to build apps that talk to AWS services directly.

AWS services used in AM/NS, Hazira

1. Amazon **EC2** — Amazon Elastic Compute Cloud
2. AWS **Lambda**
3. Amazon **S3** — Amazon Simple Storage Service
4. Amazon **DynamoDB**
5. Amazon **API Gateway**
6. AWS **IAM** — AWS Identity and Access Management
7. Amazon **CloudWatch**
8. Amazon **Bedrock**
9. AWS **Amplify**
10. Amazon **EventBridge**
11. Amazon **Textract**
12. AWS **Secrets Manager**
13. Amazon **Lex**



These services together form the backbone of AM/NS's internal cloud applications

AWS EC2 – (Compute)

Compute - Amazon EC2

Amazon Elastic Compute Cloud (EC2) is a core AWS service that provides **scalable, on-demand virtual servers** in the cloud. It allows you to rent computing capacity without managing physical hardware, using flexible pay-as-you-go pricing



Features of - EC2

Virtual servers in the cloud

User has full control over OS and software

Always-on for heavy workloads

Analogy: Renting a flat — you control the inside, AWS owns the building

Used for: Hosting backend apps, batch processing

Key note: EC2 = always on, always paying.

AWS Lambda – (Compute)

Serverless Compute - AWS Lambda

AWS Lambda is a **serverless, event-driven compute service** that allows you to run code without provisioning or managing underlying physical servers. You simply upload your code, and the service automatically handles all the infrastructure scaling, administrative maintenance, and high availability on your behalf



Features of Lambda :

Serverless — runs code without servers

Charges zero when idle

Auto-scales from 1 to millions of requests

Analogy: Sensor light — off until triggered

Used for: Event-driven tasks, API backends

Note: AWS Lambda functions have a **maximum timeout limit of 15 minutes**. If your code runs longer than 15 minutes, AWS terminates the execution.

Key note: Lambda = only runs and pays when triggered.

Amazon S3 – (Storage)



Cloud Storage - Amazon S3

Amazon S3 is a **Infinite cloud storage** for any file type — images, PDFs, videos, logs, backups

Features of S3:

99.999999999% durability

No storage limit

Files replicated across multiple data centers

Access from anywhere via URL or API

Analogy: AWS S3 is like an **infinite, highly secure digital storage locker** where every file you drop in gets its own unique, sharable web link.

AWS S3 is used to **store and access unlimited files**—like website assets, data backups, and media—securely from anywhere via the web.

DynamoDB – (Database)



NoSQL Database - Amazon DynamoDB

DynamoDB is a Fully managed NoSQL database with millisecond response times at any scale

Key features:

- No fixed schema — flexible data structure
- Single-digit millisecond performance guaranteed
- Scales automatically — no manual configuration
- Replicated across multiple Availability Zones

Real example: Netflix uses DynamoDB to load your watchlist for 200 million+ users simultaneously

Use: Employee records, equipment status, production line readings

Traditional Database	DynamoDB
Traditional SQL: Slows down under heavy traffic	Consistent speed regardless of load
complex scaling	auto-scaling
fixed schema Right	flexible structure

API Gateway

Amazon API Gateway: API Management

Amazon API Gateway is a fully managed AWS service that allows developers to **create, publish, maintain, monitor, and secure APIs** at any scale. It acts as a secure "front door" for applications to access data, business logic, or backend services like AWS Lambda, Amazon EC2, or any publicly accessible web endpoint.



Features of API Gateway :

Secure front door for all backend APIs

Handles authentication, rate limiting, routing

Hides backend from direct internet exposure

Analogy: Security check desk — every visitor gets verified before entering

Use: All employee app requests pass through API Gateway before reaching Lambda or DynamoDB

Real Example :

For booking a ride when a user opens their mobile app and taps "**Request a Ride**," the application does not talk directly to the database or individual servers. Instead, it makes a single request to **AWS API Gateway**.

EventBridge

Amazon EventBridge: Event Router

Amazon EventBridge is a fully managed, **serverless event bus service** that enables you to **build scalable event-driven applications**. It simplifies connecting your software with data from AWS services, custom apps, and third-party **SaaS platforms** by routing real-time events without needing to write complex integration code

Features of EventBridge :

Serverless event bus connecting all services

IF event happens → THEN trigger action OR
IF scheduled time arrives → THEN trigger action

Eliminates constant polling — systems react only when needed

Analogy: Smart home automation — motion detected → lights on → camera starts

Use: Document approved → notify manager + archive to S3 + update database — all automatically



IAM - (Security)

AWS Identity and Access Management (IAM): Access Control

AWS IAM (Identity and Access Management) is the foundational **security service** that **controls who is authenticated** (logged in) and **authorized** (has permissions) to use your AWS resources. It ensures the right people and apps have the right level of access to your cloud environment.

Features of IAM :

Controls who can access what in AWS

Users, groups, roles, and policies

Principle of least privilege — minimum access only

Every AWS API call is checked against IAM

Analogy: Office keycard system — each employee can only enter rooms they need

Use: HR app accesses HR data only. Production app accesses production data only. Neither can touch the other.



Secrets Manager – (Security)

AWS Secrets Manager: Credentials Vault

AWS Secrets Manager is a fully managed service that helps you securely encrypt, store, and retrieve credentials, API keys, and OAuth tokens. It eliminates the need to hardcode sensitive data into your application code by allowing you to dynamically fetch values at runtime.

Features of Secrets Manager :

Stores passwords, API keys, database credentials securely

Applications retrieve secrets at runtime — never hardcoded in code

Automatic rotation — passwords change periodically

Every access is logged

Analogy: Bank vault for passwords — only authorised apps can retrieve specific secrets

Use: Database passwords never written in code — fetched securely at runtime



Amazon CloudWatch -(Monitoring)



Monitoring & Metrics - Amazon CloudWatch

What it collects:

Logs: Text records of every event and error in every service

Metrics: Numbers — CPU usage, request count, error rate, response time

Alarms: Automatic alerts when thresholds are crossed

How it works:

- AWS services automatically send data to CloudWatch
- You build dashboards showing real-time health
- Set alarms: "If error rate > 5% for 3 minutes → send SMS to engineer"
- Set automated responses: "If CPU > 80% → add more EC2 instances automatically"

Analogy: Car dashboard — shows all vitals in real time and warns before something breaks

Use: Production systems monitored 24/7. Any service failure triggers immediate alerts to the engineering team before end users are affected

Amazon Bedrock – (AI)

Generative AI - Amazon Bedrock

Amazon Bedrock is a fully managed AWS service that provides developers with a **single API** to access and customize top-tier foundation models from leading AI companies. It enables businesses to securely build, scale, and integrate generative AI applications—such as chatbots, search tools, and autonomous AI agents—without managing underlying servers.

Features of Amazon Bedrock:

Access to multiple AI models through one API (Claude, Llama, Titan)

No model training required — plug in and use

Company data stays private — never used to train models

Build chatbots, Q&A systems, document summarizers

Analogy: A menu of AI chefs — choose which one cooks your dish



Use: Internal chatbot answering employee questions about safety procedures and HR policies using company documents

Amazon Lex – (AI)

Conversational AI - Amazon Lex

Amazon Lex is a fully managed AWS service for building conversational interfaces—like **chatbots** and **virtual agents**—using both **voice** and **text**. Powered by the same advanced deep learning engine as **Amazon Alexa**, it allows developers to easily integrate natural language capabilities into any application without needing prior artificial intelligence or machine learning expertise.



Features of Amazon Lex

- Build conversational chatbots and voice assistants
- Same technology that powers Amazon Alexa
- Understands natural language — users speak or type naturally
- Deploys to websites, mobile apps, Slack
- **Analogy:** Brain behind Alexa — embed it in your own app
- Use: HR chatbot — employees ask "How many leaves do I have?" and get instant answers without calling HR

Amazon Textract – (AI)



Intelligent Document AI - Amazon Textract

Amazon Textract is a fully managed machine learning service by AWS that **automatically extracts text, handwriting, and structured data** from scanned documents and images. It goes beyond traditional Optical Character Recognition (OCR) by understanding the context and layout of forms and tables, eliminating the need for manual data entry or custom code.

Beyond basic scanning (comparison):

Basic OCR: Just takes a photo of text

Textract: Reads the document, understands structure, extracts labeled data

What it extracts:

- Plain text from any document
- Tables with rows and columns intact
- Form fields with their labels and values
- Handwritten text alongside printed text

Workflow:

Scanned PDF → Upload to S3 → Call Textract API → Structured JSON data
→ Store in DynamoDB or send to application

Use: Supplier invoices arrive as scanned PDFs.

Textract automatically extracts invoice number, date, supplier name, line items, and total amount — feeds directly into the ERP system. Zero manual data entry.

AWS Amplify

Mobile & Frontend Full stack development - AWS Amplify

Amplify is Development platform to build and deploy **complete web** and **mobile applications** quickly — with authentication, storage, APIs, and databases set up automatically

What it handles automatically:

- User authentication and login (via Cognito)
- File storage (S3)
- Backend APIs (API Gateway + Lambda)
- Database (DynamoDB)
- Hosting and deployment



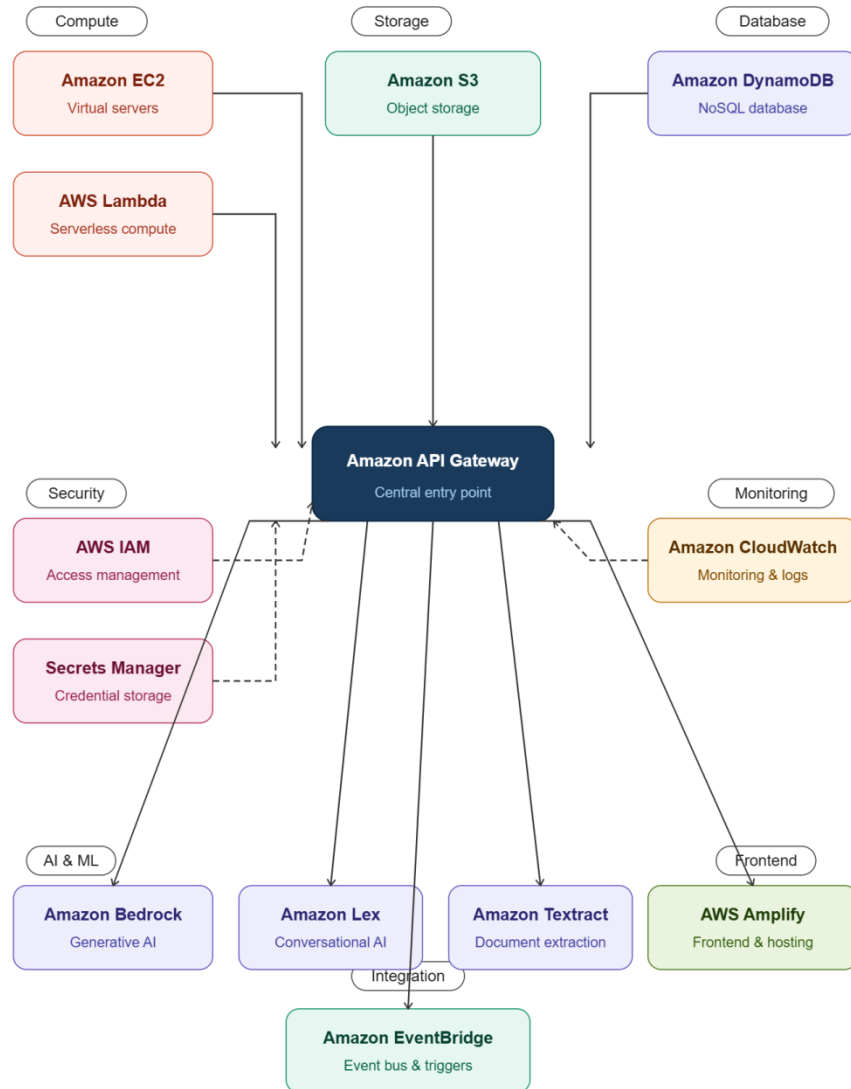
Without Amplify vs With Amplify:

Without: Configure EC2, set up API Gateway, connect DynamoDB, configure Cognito, set up S3 — each takes hours

With Amplify: Run one command per feature — everything configured and connected in minutes

Use: Internal employee portals — with login, document access, and dashboards — built and deployed using Amplify, dramatically reducing development time

Basic Workflow of AWS services



Architecture flowchart diagram with services.

What is a Vector Database?

Understanding Vector Databases — The Foundation of AI Search

A Vector Database stores data as mathematical vectors (embeddings) — lists of numbers that represent the **meaning** of data. Unlike traditional databases that match exact values, vector DBs find results by **semantic similarity** — understanding what things **mean**, not just what they say.

Traditional DB vs Vector DB

Traditional DB: Search by exact match — SQL WHERE name = 'dog'

Vector DB: Search by meaning — finds 'puppy' when you search 'dog'

How it works:

1. AI model converts text/image to a vector (e.g. [0.2, 0.8, 0.1...])
2. Vectors stored in the vector database
3. Query converted to a vector, nearest neighbors found (k-NN)

Why is it needed?

Semantic Search: Find results by intent, not keyword

AI Chatbots (RAG): AI Chatbots (RAG): Help AI find the right information before answering your question.

Recommendations: Find similar products, songs, or content

Image/Face Search: Match faces or images by visual similarity

Analogy: *Like a librarian who finds books by topic and context, not just title.*

ChromaDB: Popular open-source vector DB used locally in Python AI apps.

AWS & Vector Search

Native Vector Support vs. Integrated Vector Extensions

AWS offers **vector search capabilities** across multiple services — some **natively** built for it, others via **database extensions**.

NATIVE Vector Services

Vector search is a core feature.

Amazon OpenSearch Service

Dedicated k-NN vector engine, full-text + semantic search. Built for high-scale similarity search.

Amazon MemoryDB

In-memory vector search with Redis-compatible engine. Ultra-low latency use cases.

Amazon Bedrock Knowledge Bases

Managed RAG pipeline — auto-embeds and retrieves docs from vector stores for AI chatbots.

INTEGRATED (Extension-based)

Vector added via plugin/extension — not core

Amazon Aurora / RDS (pgvector)

PostgreSQL extension adds vector column support. Good if you already use RDS — not purpose-built.

Amazon DynamoDB

Can store vector data as attributes, but lacks native similarity search — must use external tooling.

Amazon Neptune ML

Applies vectors on relationships between data points — finds patterns like who you might know or what product you might like.

Amazon OpenSearch Service



AWS's Primary Native Vector Search Service

Amazon OpenSearch Service is a fully managed distributed search and analytics engine with a dedicated vector engine — enabling semantic search, RAG-based AI apps, and k-NN similarity search at scale.

Features of OpenSearch:

Dedicated Vector Engine: Native k-NN search — not a plugin or extension

Hybrid Search: Combine keyword + semantic search in one query

Bedrock Integration: Acts as the vector store for RAG-based AI apps

Fully Managed: No server setup — AWS handles scaling and availability

Analogy: Like a Google-grade search engine you can deploy inside your own app.

How It Works (RAG Workflow)

1. Documents uploaded to S3
2. Bedrock converts them to embeddings (vectors)
3. Vectors stored in OpenSearch vector index
4. User asks a question → converted to a vector
5. OpenSearch finds nearest matching vectors
6. Relevant docs sent to LLM → accurate answer generated

Use:

Employee queries safety manuals and HR policies via the internal Bedrock chatbot — OpenSearch retrieves the most relevant document sections to give precise answers.

Conclusion

AWS replaces expensive physical servers with flexible, on-demand digital tools.

By combining these core services in **AM/NS**, we get a complete ***ecosystem***:

Infrastructure: Compute, storage, and databases run our app smoothly.

Security & Network: Safe data roads and tight access controls lock out hackers.

Smart Features: Next-gen AI reads documents, runs chatbots, and generates content.

Ultimately, AWS allows developers to focus entirely on writing great code while Amazon handles the heavy hardware.

Thank You